Task 1: Scintillation Spectrum from Alpha Particles

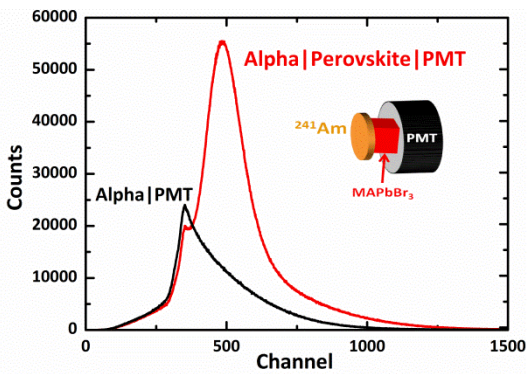

Figure 1: Alpha scintillation spectrum of MAPbBr3 single crystals. Measurement was taken at room temperature and integrated over 10 mins. *Measurement taken by Dr. Qiang Xu.*

The ultimate goal of this project is to create a system that reliably produces absolute light yield measurements. We have already been able to observe the scintillation of perovskite crystals under alpha particle exposure with a CCD camera. However we ultimately hope to quantify this scintillation by measuring the absolute light yield, or number of photons created in a material per MeV of input energy. The scintillation is very weak and therefore sensitive equipment must be used to detect these photons. An alpha particle scintillation spectrum has already been obtained and can be seen in Figure 1. This measurement will be repeated with the new experimental set up to validate the set-up. Repeating the experiment in a more stable set-up can also help to get a cleaner spectrum, possibly eliminating the alpha particles that are incident on the PMT.

However it is not plausible to obtain an absolute light yield measurement for alpha particles as their attenuation makes it difficult to identify the exact amount of energy with which they are hitting the crystal. Since this energy is most likely not the same every time, the scintillation would have to be reported as energy per alpha particle which is not a useful unit. Therefore perovskite crystals will be used to obtain an alpha particle scintillation spectrum.

Task 1.1: Detection Selection

First, the detector must be chosen. Currently we are using a Hamamatsu Model H10493-003 PMT. However the window material of this PMT has been susceptible to radiation in our previous measurements. Since alpha particles are easily attenuated they can be blocked before reaching the PMT window, minimizing the interference. More penetrating radiation sources such as X-rays and gamma rays may require another PMT with a radiation insensitive window will be necessary. [1] We have already ordered a PDM9113-CN PMT from Electron Tubes. When this arrives we will test it for our needs. It might be necessary to eschew the photosensor modules, which are PMTs prepackaged with preamplifiers and other electronics, in favor of standard PMTs. Low-light applications often require the raw photon signal to be observed. However the associated electronics of the photosensor modules amplifies thermal noise making it difficult to distinguish the low-light signal from the photons from that of thermal noise. The bare PMT could be connected to a transformer and a high voltage power supply which would allow the gain to be easily adjusted for the experiment without relying on other instruments or software.

Another option is to consider using a Silicon Photomultiplier (SiPM) which offers similar resolution to PMTs but also has easier use and more flexibility in configuration. SiPMs have been studied previously [2] and the industry is gradually replacing PMTs with SiPMs in most applications. NARS lab currently has a C8366 series photosensor amplifier with a S3399 Si PIN photodiode. However this model has a minimum detection limit at 1 Hz is about 40,000 photons at 840 nm so it is not suitable for our experiment as it cannot detect a single photon due to lack of gain in the SiPM. Another company, SensL, offers several variations of SiPMs that could also be considered.

Task 1.2: Light Tight Apparatus

Once a suitable detector is in place it is necessary to place it in a light tight set-up. This ensures that stray light will not disrupt the measurement by inducing photons that do not originate from the scintillation of the perovskite. Currently the photosensor module is held in an inverted position by a table clamp, as shown in Figure 2. The perovskite is coupled to the PMT window by silicon optical grease and then is wrapped in plumbers tape to ensure more photons are redirected to the PMT window. Aluminum foil is used to surround the crystal and PMT and then light-tight electrical tape is used to secure the foil. This creates a light-tight environment.

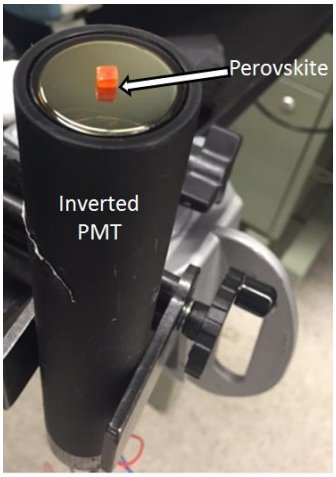

Figure 2: PMT set-up with perovskite on the window. The perovskite is coupled to the window with Teflon tape and optical grease.

However, the light tight apparatus will be improved as part of the final detection set-up. Several methods of improvement include substituting plumber's tape with Teflon tape and limiting aluminum foil to one-use to prevent creases. Use of a NEMA enclosure, designed specifically to create a light tight environment that allows for repeatable experiments, is also another option typically used in professional light tight measurements.

Task 2: Scintillation of Gamma Rays

Once the scintillation spectrum of alpha particles is completed the next step is to observe the scintillation of perovskites under gamma ray bombardment. This has already been done for alpha scintillation. The 10 μCi Na-22 in the NARS lab, which produces a 1.27 MeV gamma ray, is strong enough to induce a signal in the perovskite. While a pulse signal has been detected with the perovskite on an oscilloscope, it is weak. Therefore obtaining an image of the perovskite scintillating under gamma ray exposure will require a light-tight set up to ensure that the CCD camera can obtain a proper image.

Task 2.1: Experimental Set-Up

The CCD camera must be positioned in a way to capture an image of the glowing crystal. This must be done by placing the CCD inside of a light tight environment. The challenge is to create a setup that allows the CCD to be operated while maintaining a light tight environment. This can be done by placing the camera's controls outside of the light tight box or by putting the camera on a delay. One potential improvement would be to get a colored CCD so that the color of the perovskite scintillation could be seen.

Task 3: Absolute Light Yield of Gamma Rays Scintillation

Upon obtaining an image of the gamma scintillation, the next step will be to complete an absolute light yield measurement. First, a scintillation spectrum of the perovskite under gamma scintillation must be obtained. Using specs provided by the manufacturer, along with several other parameters, the absolute light yield can be calculated. However, it is possible that the red-sensitive ADIT tube is unable to resolve a single photoelectron peak. This means that additional steps may need to be taken to calculate the absolute light yield of the perovskite.

Task 3.1: Gamma Ray Scintillation Spectrum

Using the same set-up from Task 1, a gamma ray spectrum must be obtained. However this will have its own challenges as the highly energetic 10 μCi Na-22 source may cause signal saturation while the 1 μCi Co-60 source may be too weak to induce a signal that can be easily read and collected by the PMT. Therefore the distance between the source and crystal may need to be adjusted to create a useful signal in the PMT.

Another challenge with the experimental set-up is the system calibration. The signal from the PMT must be properly shaped to generate the full scintillation spectrum. This is done by using the amplifier to adjust the gain of the signal. The time constant of the amplifier must also be manipulated to ensure that the entire signal is collected but that there is no signal pile-up. Additionally the pole-zero cancelation must be properly tuned to ensure that no overshoot or undershoot disrupts the signal.

Task 3.2 Absolute Light Yield Calculation

Once the gamma scintillation spectrum is obtained then the absolute light yield of the perovskite can be calculated. This is done by using the PMT in a light-tight environment with no source or crystal to observe the channel number of a thermally induced photoelectron at high gain. The radioluminescence spectrum, observed with a spectrometer when the perovskite is excited by X-rays or Gamma-rays, is also taken. The channel number and gain for the full energy peak of the scintillator is then recorded. These results are used to calculate the number of scintillation photoelectrons by dividing the full energy peak of the scintillator by the single photoelectron peak of the PMT. Once the number of photoelectrons is known it is multiplied by the integrated spectrum of the PMT quantum efficiency to get the number of photons created per MeV of energy. The resulting equation is shown in Figure 3.

$$N_{phe}(E) \propto N_{ph}(E) \times \eta_L \times Q.E.(\lambda) \times \varepsilon$$

where:

$N_{ph}(E)$ - number of photons per energy unit produced in the scintillator at a given energy,

η_L - light collection efficiency,

$Q.E.(\lambda)$ - quantum efficiency,

ε - efficiency of photoelectron or electron-hole pairs collection.

Figure 3: Absolute Light Yield Calculation, taken from [3].

Task 3.3: Absolute Light Yield with a Reference Crystal

Validation of the measurement will require a reference crystal whose absolute light yield is already known. NaI, BGO and LYSO are all common gamma scintillation crystals. We have ordered a BGO crystal because it is gamma sensitive and its emission spectrum peaks around 480 nm but also extends weakly into the red region which helps when comparing it to the red-photon emissions of the perovskite. Besides system verification the reference crystal may also be needed in the event that the red-sensitive PMT cannot resolve a single photoelectron peak. The method in Task 3.2 will be used to determine the absolute light yield of the BGO crystal with a high quantum efficiency PMT. The photopeak channel number of the BGO on the red-sensitive PMT will be divided into the photopeak channel number of the perovskite on the high quantum efficiency PMT, which can resolve the single photoelectron peak. The integral quantum efficiency of the reference crystal with the ADIT PMT will be divided by the integral quantum efficiency of the perovskite coupled to the other PMT. These two fractions will be multiplied

with the absolute light yield of the BGO with the ADIT tube to get the absolute light yield of the perovskite. The calculation is therefore

$$ALY_{perovskite} = ALY_{BGO} * \frac{Channel_{perovskite}}{Channel_{BGO}} * \frac{QE_{BGO}}{QE_{perovskite}}$$

where $ALY_{perovskite}$ is the absolute light yield of the perovskite, ALY_{BGO} is the absolute light yield of the BGO crystal measured on a high quantum efficiency PMT. $Channel_{perovskite}$ is the channel of the photopeak on the high quantum efficiency PMT, $Channel_{BGO}$ is the channel of the photopeak on the ADIT PMT, QE_{BGO} is the integral quantum efficiency of the BGO on the ADIT PMT and $QE_{perovskite}$ is the integral quantum efficiency of the perovskite on the ADIT PMT.

References

[1] Hamamatsu, Photomultiplier Tube Handbook, Edition 3a, 2007.

[2] de Haas, Johan TM, and Pieter Dorenbos. "Advances in yield calibration of scintillators." IEEE Transactions on Nuclear Science 55.3 (2008): 1086-1092.

[3] Moszynski, M., et al. "Absolute light output of scintillators." IEEE transactions on nuclear science 44.3 (1997): 1052-1061.